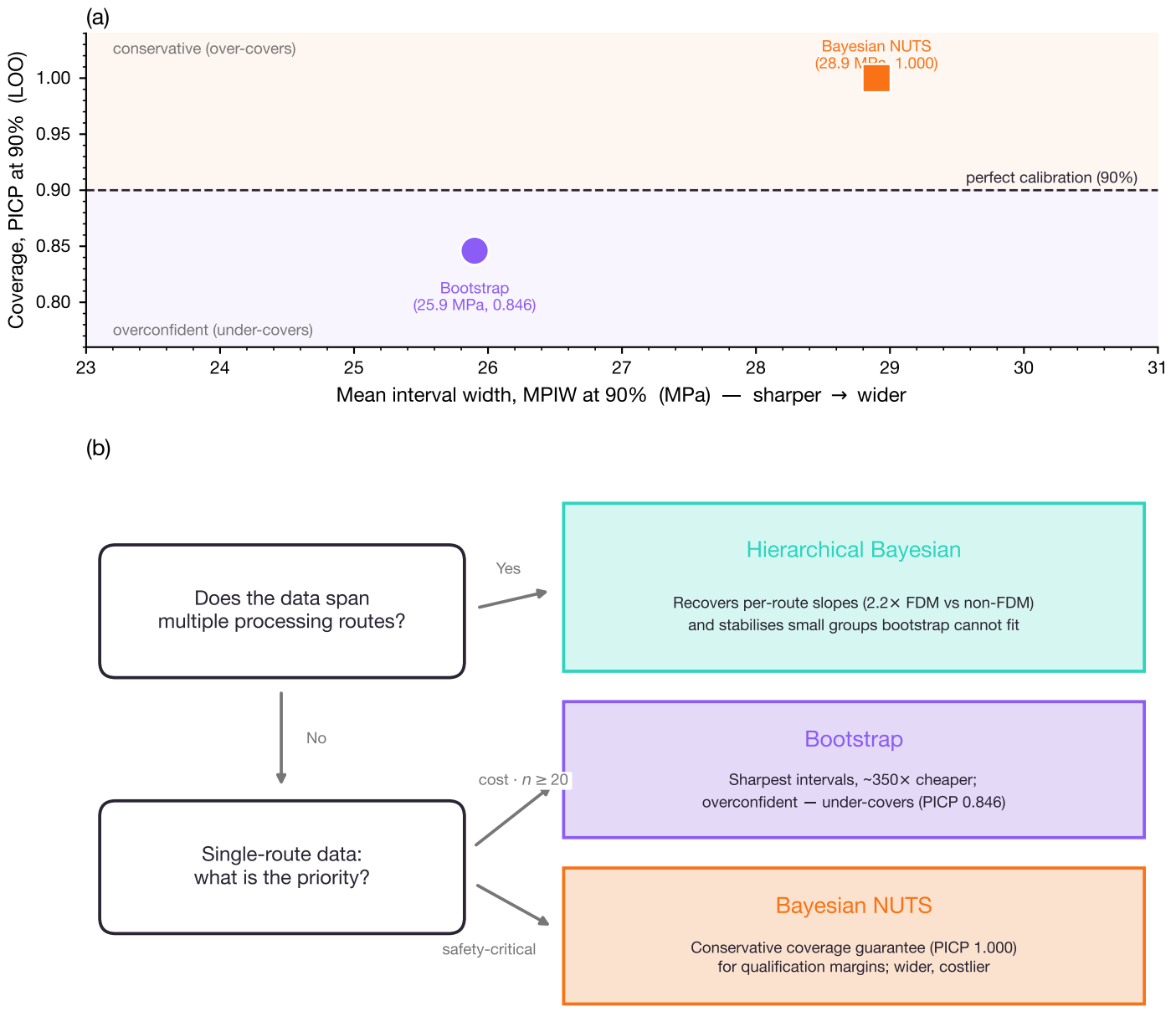
(a)
conservative (over-covers)
Bayesian NUTS
(28.9 MPa, 1.000)
perfect calibration (90%)
Bootstrap
(25.9 MPa, 0.846)
overconfident (under-covers)
Coverage, PICP at 90% (LOO)
1.00
0.95
0.90
0.85
0.80
23
24
25
26
27
28
29
30
31
Mean interval width, MPIW at 90% (MPa) — sharper → wider
(b)
Does the data span multiple processing routes?
Yes
Hierarchical Bayesian
Recovers per-route slopes (2.2× FDM vs non-FDM) and stabilises small groups bootstrap cannot fit
No
Single-route data: what is the priority?
cost · n ≥ 20
Bootstrap
Sharpest intervals, ~350× cheaper; overconfident — under-covers (PICP 0.846)
safety-critical
Bayesian NUTS
Conservative coverage guarantee (PICP 1.000) for qualification margins; wider, costlier